

SKILL WEEK - 2

1. Task -1:

```
[(base) saiabburi@Geyasree-MacBook-Air ~ % cd ~/OSSP
[(base) saiabburi@Geyasree-MacBook-Air OSSP % nano fork_exec.c
```

```
[(base) saiabburi@Geyasree-MacBook-Air OSSP % gcc fork_exec.c -o fork_exec
[(base) saiabburi@Geyasree-MacBook-Air OSSP % ./fork_exec
Enter a Linux command: ls

=== Parent Process ===
Parent PID : 3819
Child PID : 3832

=== Child Process ===
Child PID : 3832
Parent PID : 3819
Executing command: ls

LICENSE      README.md   bin          fork_exec   include     scripts     tests
Makefile     assets     docs         fork_exec.c obj         src

Child process completed.
[(base) saiabburi@Geyasree-MacBook-Air OSSP %
```

```
[(base) saiabburi@Geyasree-MacBook-Air OSSP % tree
```

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── fork_exec
├── fork_exec.c
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c
```

```
9 directories, 20 files
```

```
[(base) saiabburi@Geyasree-MacBook-Air OSSP %
```

2. Task -1:

```
[(base) saiabburi@Geyasree-MacBook-Air OSSP % man fork
```

```
Unknown locale, assuming C
```

```
[(base) saiabburi@Geyasree-MacBook-Air OSSP % man exec
```

```
Unknown locale, assuming C
```

```
[(base) saiabburi@Geyasree-MacBook-Air OSSP %
```

```
[(base) saiabburi@Geyasree-MacBook-Air OSSP % man execvp
```

```
Unknown locale, assuming C
```

Task -2:

```
/ directories, 20 files  
[(base) saiabburi@Geyasree-MacBook-Air OSSP % nano src/main.c
```

```
UW PICO 5.09
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    pid_t pid;
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if(pid == 0)
    {
        printf("\n=== Child Process ===\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
        printf("Executing command: %s\n\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed.\n");
    }
    else if(pid > 0)
    {
        printf("\n=== Parent Process ===\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }
    else
    {
        printf("Fork failed!\n");
    }

    return 0;
}
```

```
[(base) saiabburi@Geyasree-MacBook-Air OSSP % ls bin
my_shell
[(base) saiabburi@Geyasree-MacBook-Air OSSP % ls
LICENSE          assets          fork_exec        obj              tests
Makefile         bin            fork_exec.c      scripts
README.md        docs          include          src
[(base) saiabburi@Geyasree-MacBook-Air OSSP % ./bin/my_shell
Enter a Linux command: ls

=== Parent Process ===
Parent PID : 5678
Child PID : 5693

=== Child Process ===
Child PID : 5693
Parent PID : 5678
Executing command: ls

LICENSE          assets          fork_exec        obj              tests
Makefile         bin            fork_exec.c      scripts
README.md        docs          include          src

Child process completed.
```